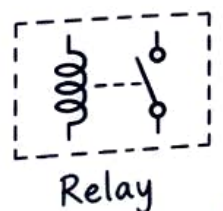
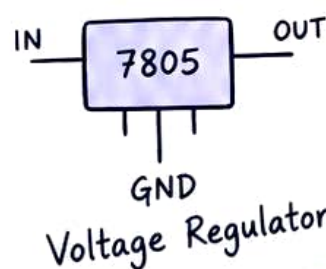
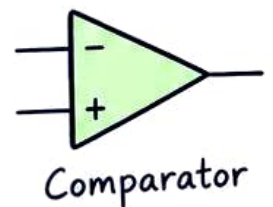
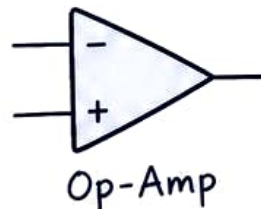
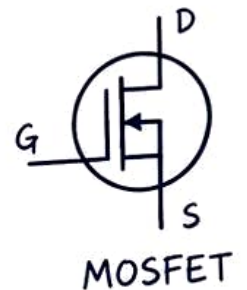
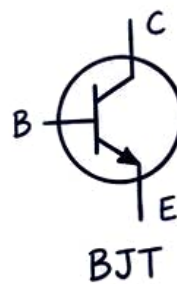
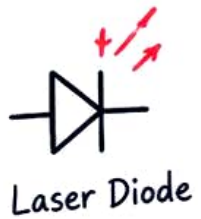
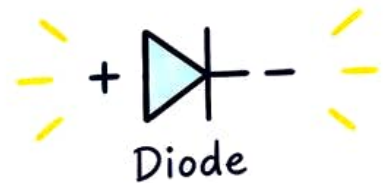


Semiconductor & Components

From Diodes to Logic —
The Building Blocks of Electronics

Topics Covered

- 1 Diode
- 2 Zener Diode
- 3 LED, Photodiode & Laser Diode
- 4 Transistor — BJT
- 5 MOSFET — How It Works
- 6 BJT vs MOSFET
- 7 Op-Amp Fundamentals
- 8 Comparator vs Op-Amp
- 9 Voltage Regulators
- 10 Relay vs Transistor Switching



Small Components...
Big Impact!

Diode

A diode is a two-terminal semiconductor device that allows current to flow in one direction only.

Working

- Forward Bias → low resistance (current flows)
- Reverse Bias → high resistance (almost no current)
- Acts as a one-way valve for current.

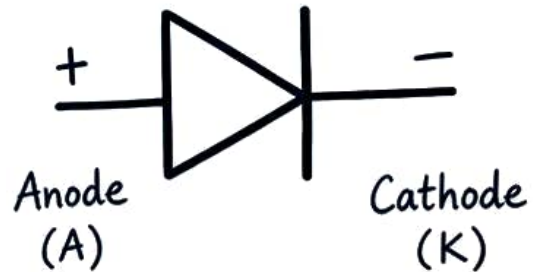
Applications

- ✓ Rectification (AC → DC)
- ✓ Protection (e.g. reverse polarity, flyback diode)
- ✓ Signal demodulation
- ✓ Clipping & clamping circuits
- ✓ Switching applications

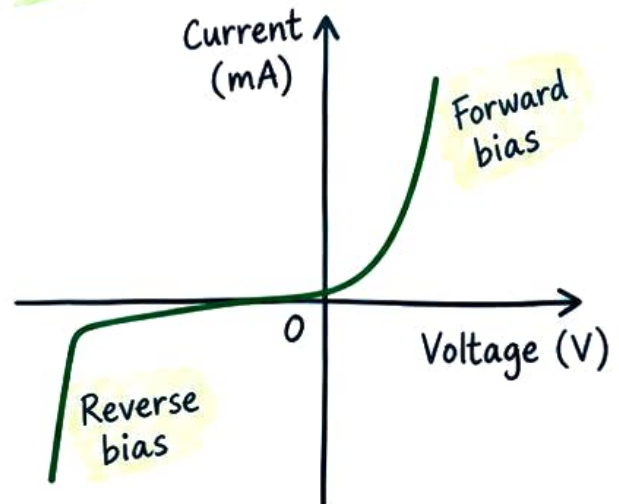


Small device...
Big role in electronics!

Symbol



V-I Characteristic



Different types
of diodes exist
(for different needs).

Zener Diode

A Zener diode is a specially designed diode that works in reverse bias region to provide a stable reference voltage.

Working

- In forward bias, it behaves like a normal diode (conducts after threshold voltage).
- In reverse bias, when the voltage reaches the Zener voltage (V_Z), it conducts heavily while keeping the voltage almost constant.
- This is called the Zener breakdown (used for voltage regulation).

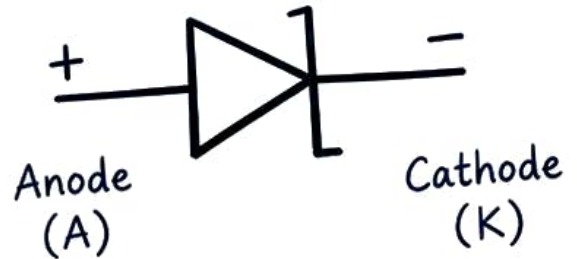
Applications

- ✓ Voltage regulation (fixed reference voltage)
- ✓ Overvoltage protection
- ✓ Used in clippers and clampers
- ✓ Wave shaping and measurement circuits

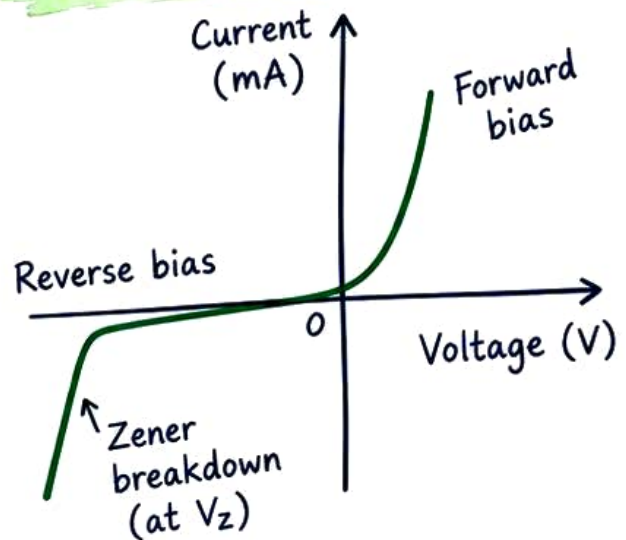


Provides stable reference voltage!

Symbol



V-I Characteristic



Key Point

Zener diode works in the reverse breakdown region, not the normal reverse region.

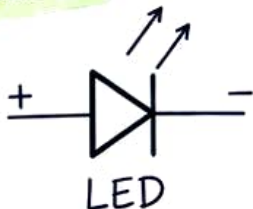
LED, Photodiode & Laser Diode

Three light-based semiconductor devices, used for different purposes — emitting, detecting and amplifying light.

LED

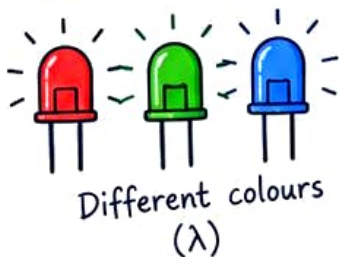
- Light Emitting Diode.
- Emits light when forward biased (electrons recombine with holes in the junction).
- Available in different colours (wavelengths).

Symbol



Applications

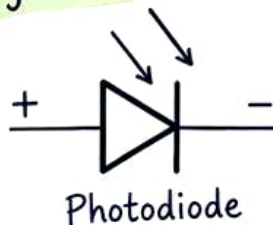
- Indicators (status lights)
- Displays
- Lighting
- Optical communication



Photodiode

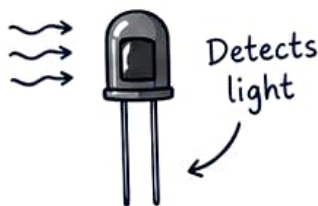
- Light sensitive diode.
- Generates current when light falls on it (photovoltaic effect).
- Works in reverse bias (for better response).

Symbol



Applications

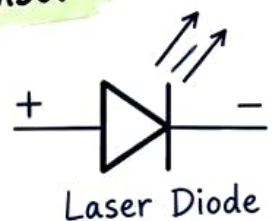
- Light sensors
- IR receivers
- Optical communication
- Position and speed sensing



Laser Diode

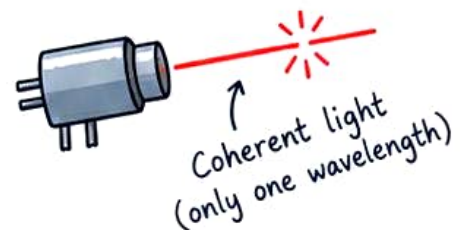
- Emits coherent and monochromatic light.
- Works in forward bias (with optical feedback from cavity).
- High light intensity and directionality.

Symbol



Applications

- Fiber optic communication
- Barcode scanners
- CD/DVD players
- Medical & industrial lasers



Same semiconductor family...
Different light behaviour!



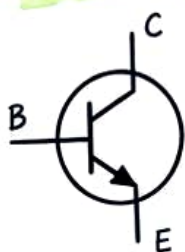
Transistor - BJT

A Bipolar Junction Transistor (BJT) is a three-terminal device used for amplification and switching.

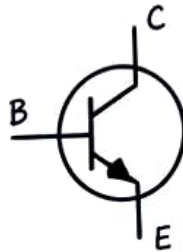
What is BJT?

- Uses both electrons and holes (bipolar).
- Has three regions: Emitter, Base and Collector.
- Available in two types: NPN and PNP.

Symbol

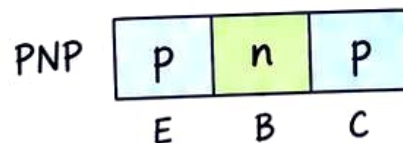
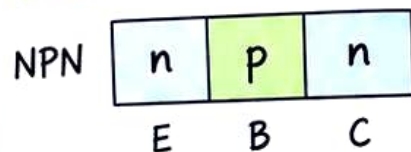


NPN
(arrow out)



PNP
(arrow in)

Structure



Working (Active Region)

- Emitter → heavily doped (injects carriers).
- Base → very thin and lightly doped (less recombination).
- Collector → moderately doped (collects carriers).
- When V_{BE} is forward biased and V_{CB} is reverse biased, current flows from emitter to collector.

Modes of Operation

- 1 Active Region (amplification)
- 2 Saturation (both junctions forward biased)
- 3 Cut-off (both junctions reverse biased)

Key Characteristics

- Current gain (β) = $\frac{I_C}{I_B}$
- High input impedance (especially compared to FET).
- Used in amplifiers, switches, oscillators, digital circuits.

Key Formulas

$$I_E = I_C + I_B$$

$$I_C = \beta I_B$$

$$I_B = \frac{I_C}{\beta}$$

(β = current gain)

Applications

- ✓ Signal amplification
- ✓ Switching circuits
- ✓ Oscillators
- ✓ Digital logic (TTL, ECL)



Small device...
Big role in electronics!



MOSFET

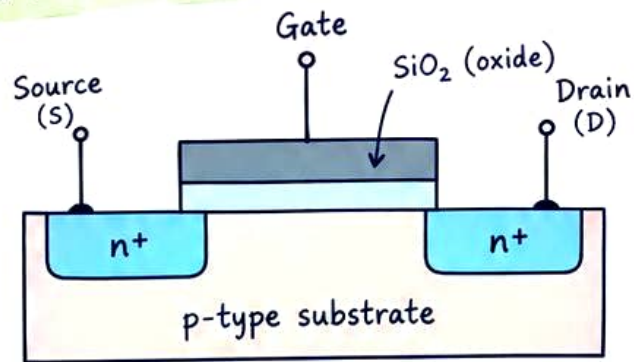
How It Works

A MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor) is a voltage-controlled device used for switching and amplification.

What is MOSFET?

- Uses a gate voltage to control the current between source and drain.
- Very high input impedance (less current required).
- Available in two types: N-channel and P-channel.

Structure (N-channel Enhancement Type)



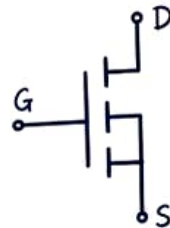
Gate is insulated by a thin oxide layer, so almost no current flows into the gate.

How It Works

- 1 $V_{GS} < V_{TH}$ → No channel (cut-off).
- 2 $V_{GS} > V_{TH}$ → Inversion layer forms (channel created).
- 3 V_{DS} applied → Electrons flow from source to drain (current flows).

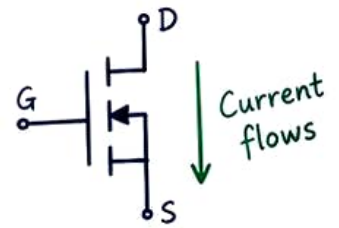
Operation Modes (N-channel)

Cut-off ($V_{GS} < V_{TH}$)



No current flows.

Linear / Triode ($V_{GS} > V_{TH}$, small V_{DS})



Acts like a resistor.

Key Points

- ✓ Voltage-controlled device
- ✓ High input impedance
- ✓ Fast switching
- ✓ Low power consumption



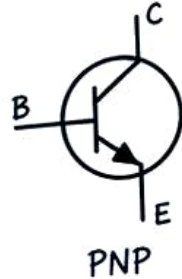
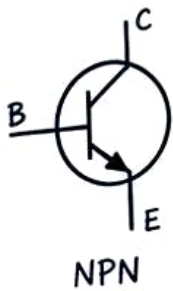
Used in digital circuits, power electronics and modern control systems.

BJT vs MOSFET

Both are semiconductor devices, but they work differently and have different strengths.

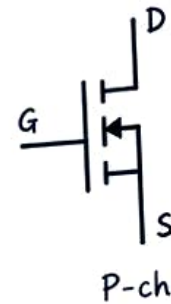
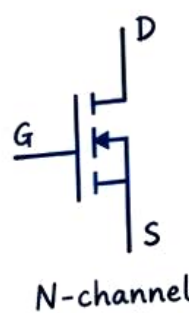
BJT (Bipolar Junction Transistor)

- Current controlled device (base current controls collector current).
- Uses both electrons and holes (bipolar).
- Has 3 terminals: Emitter, Base, Collector.
- Available in NPN and PNP types.



MOSFET (Metal-Oxide-Semiconductor FET)

- Voltage controlled device (gate voltage controls current).
- Uses only one type of carrier (unipolar).
- Has 3 terminals: Gate, Drain, Source.
- Available in N-channel and P-channel.



(Arrow points out for N-channel, in for P-channel)

Quick Comparison

Feature	BJT	MOSFET
• Control	Current controlled	Voltage controlled
• Carriers	Bipolar (electrons + holes)	Unipolar (single carrier)
• Input impedance	Low to medium	Very high
• Switching speed	Slower	Faster
• Power consumption	Higher (due to base current)	Lower (almost no gate current)
• Thermal behaviour	More prone to thermal runaway	Better thermal stability
• Cost (general)	Usually lower (for low power)	Higher (for same rating)

Key Takeaway



Use BJT when you need high current gain and are working with low-frequency or analog circuits.

Use MOSFET when you need high input impedance, fast switching and low power consumption.

Op Amp Fundamentals

An operational amplifier (op amp) is a high-gain, differential amplifier with very high input impedance and low output impedance.

Working

- It amplifies the difference between two input voltages (V_+ and V_-).
- The output voltage changes based on the input difference and the open-loop gain (very high).
- It can work with negative feedback to give stable and useful output.

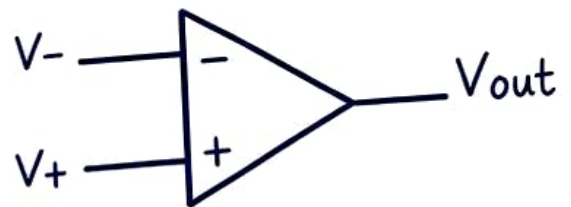
Applications

- ✓ Signal amplification
- ✓ Comparator circuits
- ✓ Filtering (active filters)
- ✓ Oscillators and waveform generation
- ✓ Sensor and measurement systems



Provides high gain & accurate output!

Symbol



Key Characteristics

- Very high input impedance
- Very low output impedance
- High open-loop gain
- Wide frequency response
- High common-mode rejection ratio (CMRR)

Key Point

Op amp doesn't produce output on its own, it needs a power supply and feedback for proper operation.

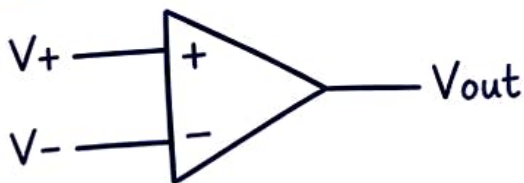
Comparator vs Op-Amp

Both are used for comparing voltages, but they work differently and are designed for different purposes.

Comparator

- Designed to compare two input voltages and give a digital output (high or low).
- Has no negative feedback.
- Works in open-loop configuration (very high gain).
- Used in threshold detection, zero-crossing, level detection, etc.

Symbol



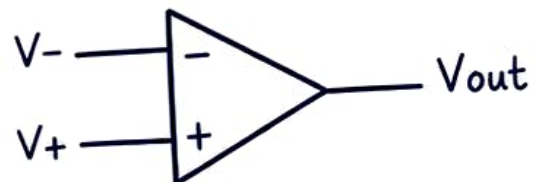
Key Points

- Digital output (saturation levels)
- Very high gain
- No feedback
- Fast response

Operational Amplifier

- Designed for signal amplification and mathematical operations (addition, subtraction, integration, etc.).
- Uses negative feedback.
- Works in closed-loop (with feedback) for stable and controlled gain.
- Used in filters, amplifiers, integrators, differentiators, etc.

Symbol



Key Points

- Analog output (continuous signal)
- Variable gain (with feedback)
- High input impedance
- Wide bandwidth



Same symbol, different jobs!



Voltage Regulators

A voltage regulator is a circuit or device that maintains a constant output voltage even when the input voltage or load changes.

Types

- Linear Voltage Regulator (e.g., 78xx, 79xx series)
- Switching Voltage Regulator (e.g., buck, boost, buck-boost)
- Adjustable Voltage Regulator (e.g., LM317)

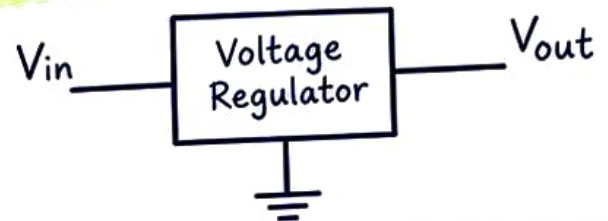
Working

- Compares the output voltage with a reference voltage.
- Adjusts the pass element (or switching action) to keep the output constant.
- Provides regulation against changes in input voltage, load current and temperature.

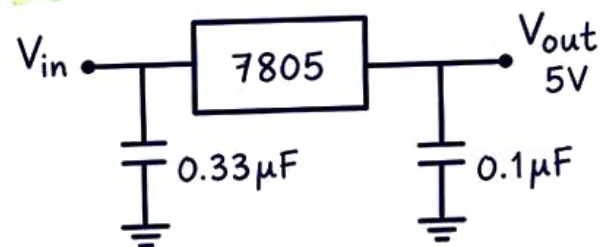
Applications

- ✓ Power supplies (electronic devices)
- ✓ Microcontrollers & logic circuits
- ✓ Communication systems
- ✓ Battery charging circuits
- ✓ Industrial control systems

Symbol



Example: 78xx (Fixed Linear)



Key Points

- Improves stability and accuracy of output voltage.
- Reduces ripple and noise.
- Has a defined output voltage (e.g., 5V, 12V).
- Can be fixed or adjustable.
- Linear regulators are simple but less efficient; switching regulators are more efficient.

Key Point

A voltage regulator keeps the output voltage constant, no matter what happens at the input or load.



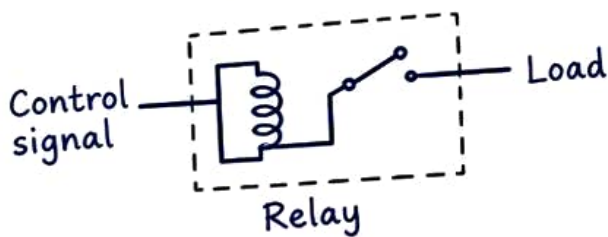
Stable voltage
for reliable circuits!

Relay vs Transistor Switching

Both relay and transistor can be used as switches, but they work in different ways and are suited for different applications.

Relay

- Uses an electromagnetic coil to operate a mechanical switch (contacts).
- Provides isolation between control and load.
- Can handle high voltage and high current.
- Slower switching speed (ms range).



Applications

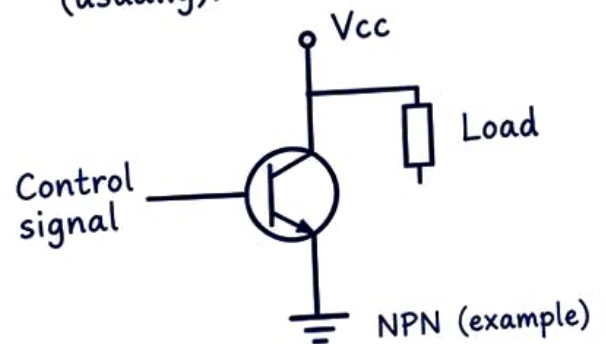
- ✓ Relay: high power systems, industrial control, isolation needed.
- ✓ Transistor: low power electronics, fast switching, digital circuits.



Choose the right switch for the right job!

Transistor

- Uses semiconductor material (e.g., BJT or MOSFET).
- No mechanical parts.
- Faster switching speed (ns to μ s range).
- Lower power consumption.
- Cannot provide isolation (usually).



Key Point

Relay = mechanical, high power, slower, isolated.

Transistor = electronic, low power, faster, no isolation.

